

Lecture 17: Asymptotics

Economics 326 — Introduction to Econometrics II

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Why we need large-sample theory

- The OLS estimator $\hat{\beta}$ has desirable properties:
 - $\hat{\beta}$ is unbiased if the errors are **strongly exogenous**: $E[U_i | \mathbf{X}] = 0$.
 - If in addition the errors are **homoskedastic**, then $\widehat{\text{Var}}(\hat{\beta}) = s^2 / \sum_{i=1}^n (X_i - \bar{X})^2$ is an unbiased estimator of the conditional variance of $\hat{\beta}$.
 - If in addition the errors are **normally** distributed (given $\mathbf{X}$), then $T = (\hat{\beta} - \beta) / \sqrt{\widehat{\text{Var}}(\hat{\beta})}$ has a t distribution which can be used for hypothesis testing.

Limitations of finite-sample theory

- If the errors are only **weakly exogenous**:

$$E[X_i U_i] = 0,$$

the OLS estimator is in general biased.

- If the errors are **heteroskedastic**:

$$E[U_i^2 | X_i] = h(X_i),$$

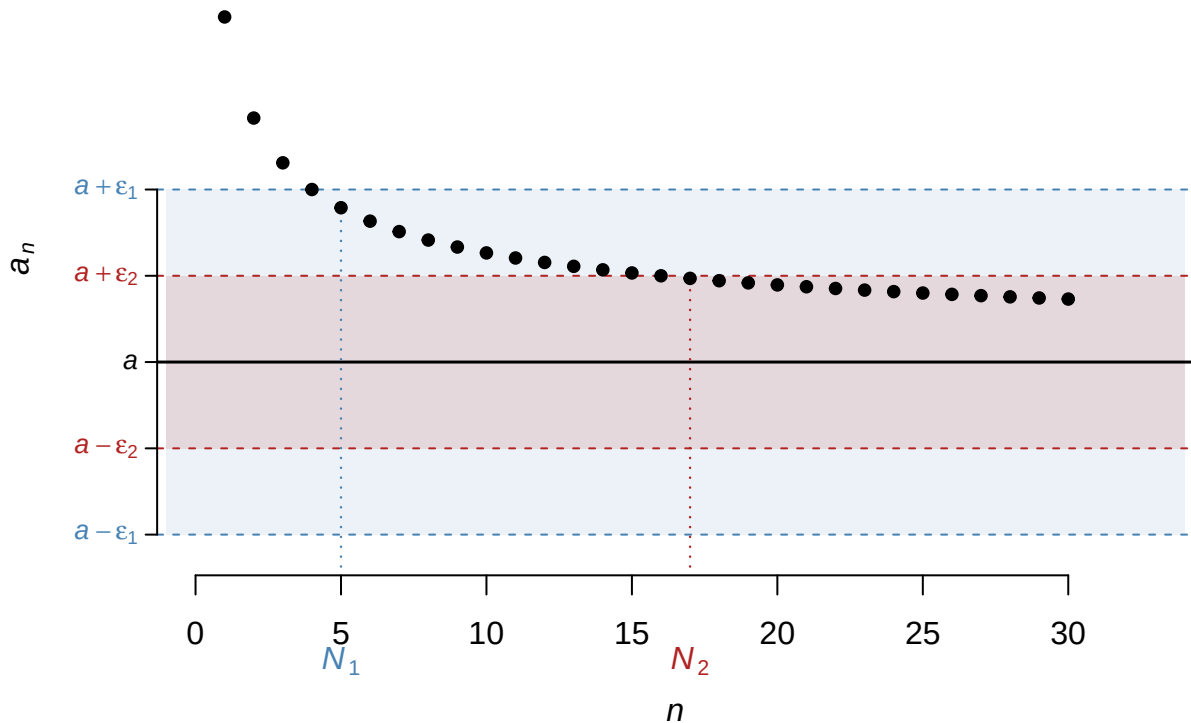
the “usual” variance formula is invalid; we also do not have an unbiased estimator for the variance in this case.

- If the errors are **not normally distributed** conditional on $\mathbf{X}$, then T - and F -statistics do not have t and F distributions under the null hypothesis.
- Asymptotic (large-sample) theory allows us to derive **approximate** properties and distributions of estimators and test statistics by assuming that the sample size n is very large.

Part I: Consistency

Convergence of a sequence

- A sequence of real numbers $a_1, a_2, \dots$ **converges** to a if for every $\varepsilon > 0$ there exists N such that $|a_n - a| < \varepsilon$ for all $n \geq N$. We write $a_n \rightarrow a$.



Since $\varepsilon_1 > \varepsilon_2$, the ε_2 -band is narrower, so it takes more terms for the sequence to stay inside it: $N_2 > N_1$. Smaller ε requires larger N .

- Our estimator $\hat{\beta}_n$ is **random**: its value changes with each sample. To apply the concept of convergence, we need to convert it into a **non-random** sequence indexed by n .
- We take $a_n = P(|\hat{\beta}_n - \beta| \geq \varepsilon)$, which is a non-random number for each n . We say $\hat{\beta}_n$ **converges in probability** to β if $a_n \rightarrow 0$ for all $\varepsilon > 0$.

Convergence in probability and LLN

- More generally, let θ_n be a sequence of random variables indexed by the sample size n . We say that θ_n **converges in probability** to θ if

$$\lim_{n \rightarrow \infty} P(|\theta_n - \theta| \geq \varepsilon) = 0 \text{ for all } \varepsilon > 0.$$

- We denote this as $\theta_n \rightarrow_p \theta$ or $p \lim \theta_n = \theta$.
- An example of convergence in probability is a Law of Large Numbers (LLN):

Let $X_1, X_2, \dots, X_n$ be a random sample such that $E[X_i] = \mu$ for all $i = 1, \dots, n$, and define $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$. Then, under certain conditions,

$$\bar{X}_n \rightarrow_p \mu.$$

LLN

- Let $X_1, \dots, X_n$ be a sample of **independent identically distributed (iid)** random variables. Let $E[X_i] = \mu$. If $\text{Var}(X_i) = \sigma^2 < \infty$, then

$$\bar{X}_n \rightarrow_p \mu.$$

- In fact when the data are **iid**, the LLN holds if

$$E[|X_i|] < \infty,$$

but we prove the result under a stronger assumption that $\text{Var}(X_i) < \infty$.

Markov's inequality

- **Markov's inequality.** Let W be a random variable. For $\varepsilon > 0$ and $r > 0$,

$$P(|W| \geq \varepsilon) \leq \frac{E[|W|^r]}{\varepsilon^r}.$$

- With $r = 2$, we have **Chebyshev's inequality**. Suppose that $E[X] = \mu$. Take $W \equiv X - \mu$ and apply Markov's inequality with $r = 2$. For $\varepsilon > 0$,

$$\begin{aligned} P(|X - \mu| \geq \varepsilon) &\leq \frac{E[|X - \mu|^2]}{\varepsilon^2} \\ &= \frac{\text{Var}(X)}{\varepsilon^2}. \end{aligned}$$

- The probability of observing an outlier (a large deviation of X from its mean μ) can be bounded by the variance.

Proof of Markov's inequality

- For any event A , the expectation of its indicator equals the probability of the event:

$$E[\mathbf{1}(A)] = 1 \cdot P(A) + 0 \cdot P(A^c) = P(A).$$

- Define the indicator $\mathbf{1}(|W| \geq \varepsilon)$, which equals 1 when $|W| \geq \varepsilon$ and 0 otherwise. Then:

$$\begin{aligned} \mathbf{1}(|W| \geq \varepsilon) &= \mathbf{1}(|W|^r \geq \varepsilon^r) \\ &= \mathbf{1}\left(\frac{|W|^r}{\varepsilon^r} \geq 1\right) \\ &\leq \frac{|W|^r}{\varepsilon^r} \\ \Rightarrow \\ P(|W| \geq \varepsilon) &= E[\mathbf{1}(|W| \geq \varepsilon)] \leq \frac{E[|W|^r]}{\varepsilon^r}. \end{aligned}$$

Proof of the LLN

- Fix $\varepsilon > 0$ and apply Markov's inequality with $r = 2$:

$$\begin{aligned} P(|\bar{X}_n - \mu| \geq \varepsilon) &= P\left(\left|\frac{1}{n} \sum_{i=1}^n X_i - \mu\right| \geq \varepsilon\right) \\ &= P\left(\left|\frac{1}{n} \sum_{i=1}^n (X_i - \mu)\right| \geq \varepsilon\right) \\ &\leq \frac{E\left[\left(\frac{1}{n} \sum_{i=1}^n (X_i - \mu)\right)^2\right]}{\varepsilon^2} \\ &= \frac{1}{n^2 \varepsilon^2} \left(\sum_{i=1}^n E[(X_i - \mu)^2] + \sum_{i=1}^n \sum_{j \neq i}^n E[(X_i - \mu)(X_j - \mu)] \right) \\ &= \frac{1}{n^2 \varepsilon^2} \left(\sum_{i=1}^n \text{Var}(X_i) + \sum_{i=1}^n \sum_{j \neq i}^n \text{Cov}(X_i, X_j) \right) \\ &= \frac{n\sigma^2}{n^2 \varepsilon^2} = \frac{\sigma^2}{n \varepsilon^2} \rightarrow 0 \text{ as } n \rightarrow \infty \text{ for all } \varepsilon > 0. \end{aligned}$$

Averaging and variance reduction

- Let $X_1, \dots, X_n$ be a sample and suppose that

$$\begin{aligned}E[X_i] &= \mu \text{ for all } i = 1, \dots, n, \\ \text{Var}(X_i) &= \sigma^2 \text{ for all } i = 1, \dots, n, \\ \text{Cov}(X_i, X_j) &= 0 \text{ for all } j \neq i.\end{aligned}$$

- The mean of the sample average:

$$\begin{aligned}E[\bar{X}_n] &= E\left[\frac{1}{n} \sum_{i=1}^n X_i\right] \\ &= \frac{1}{n} \sum_{i=1}^n E[X_i] \\ &= \frac{1}{n} \sum_{i=1}^n \mu = \frac{1}{n} n\mu = \mu.\end{aligned}$$

Variance of the sample average

- The variance of the sample average:

$$\begin{aligned}\text{Var}(\bar{X}_n) &= \text{Var}\left(\frac{1}{n} \sum_{i=1}^n X_i\right) \\ &= \frac{1}{n^2} \text{Var}\left(\sum_{i=1}^n X_i\right) \\ &= \frac{1}{n^2} \left(\sum_{i=1}^n \text{Var}(X_i) + \sum_{i=1}^n \sum_{j \neq i}^n \text{Cov}(X_i, X_j) \right) \\ &= \frac{1}{n^2} \left(\sum_{i=1}^n \sigma^2 + \sum_{i=1}^n \sum_{j \neq i}^n 0 \right) \\ &= \frac{1}{n^2} n\sigma^2 = \frac{\sigma^2}{n}.\end{aligned}$$

- The variance of the average approaches zero as $n \rightarrow \infty$ if the observations are **uncorrelated**.

Convergence in probability: properties

- Slutsky's Lemma.** Suppose that $\theta_n \rightarrow_p \theta$, and let g be a function continuous at θ . Then,

$$g(\theta_n) \rightarrow_p g(\theta).$$

- If $\theta_n \rightarrow_p \theta$, then $\theta_n^2 \rightarrow_p \theta^2$.
- If $\theta_n \rightarrow_p \theta$ and $\theta \neq 0$, then $1/\theta_n \rightarrow_p 1/\theta$.

- Suppose that $\theta_n \rightarrow_p \theta$ and $\lambda_n \rightarrow_p \lambda$. Then,

- $\theta_n + \lambda_n \rightarrow_p \theta + \lambda$.
- $\theta_n \lambda_n \rightarrow_p \theta \lambda$.
- $\theta_n / \lambda_n \rightarrow_p \theta / \lambda$ provided that $\lambda \neq 0$.

Consistency

- Let $\hat{\beta}_n$ be an estimator of β based on a sample of size n .
- We say that $\hat{\beta}_n$ is a **consistent** estimator of β if as $n \rightarrow \infty$,

$$\hat{\beta}_n \rightarrow_p \beta.$$

- Consistency means that the **probability** of the event that the distance between $\hat{\beta}_n$ and β exceeds $\varepsilon > 0$ can be made arbitrarily small by increasing the sample size.

Consistency of OLS

- Suppose that:
 1. The data $\{(Y_i, X_i) : i = 1, \dots, n\}$ are iid.
 2. $Y_i = \beta_0 + \beta_1 X_i + U_i$, where $E[U_i] = 0$.
 3. $E[X_i U_i] = 0$.
 4. $0 < \text{Var}(X_i) < \infty$.
- Let $\hat{\beta}_{0,n}$ and $\hat{\beta}_{1,n}$ be the OLS estimators of β_0 and β_1 based on a sample of size n . Under Assumptions 1–4,

$$\hat{\beta}_{0,n} \rightarrow_p \beta_0,$$

$$\hat{\beta}_{1,n} \rightarrow_p \beta_1.$$

- The key identifying assumption is Assumption 3: $\text{Cov}(X_i, U_i) = 0$.

Proof of consistency

- Write

$$\begin{aligned} \hat{\beta}_{1,n} &= \frac{\sum_{i=1}^n (X_i - \bar{X}_n) Y_i}{\sum_{i=1}^n (X_i - \bar{X}_n)^2} = \beta_1 + \frac{\sum_{i=1}^n (X_i - \bar{X}_n) U_i}{\sum_{i=1}^n (X_i - \bar{X}_n)^2} \\ &= \beta_1 + \frac{\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X}_n) U_i}{\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X}_n)^2}. \end{aligned}$$

- We will show that

$$\begin{aligned} \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X}_n) U_i &\rightarrow_p 0, \\ \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X}_n)^2 &\rightarrow_p \text{Var}(X_i), \end{aligned}$$

- Since $\text{Var}(X_i) \neq 0$,

$$\hat{\beta}_{1,n} = \beta_1 + \frac{\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X}_n) U_i}{\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X}_n)^2} \rightarrow_p \beta_1 + \frac{0}{\text{Var}(X_i)} = \beta_1.$$

Numerator converges to zero

$$\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X}_n) U_i = \frac{1}{n} \sum_{i=1}^n X_i U_i - \bar{X}_n \left(\frac{1}{n} \sum_{i=1}^n U_i \right).$$

By the LLN,

$$\begin{aligned} \frac{1}{n} \sum_{i=1}^n X_i U_i &\rightarrow_p \mathbb{E}[X_i U_i] = 0, \\ \bar{X}_n &\rightarrow_p \mathbb{E}[X_i], \\ \frac{1}{n} \sum_{i=1}^n U_i &\rightarrow_p \mathbb{E}[U_i] = 0. \end{aligned}$$

Hence,

$$\begin{aligned} \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X}_n) U_i &= \frac{1}{n} \sum_{i=1}^n X_i U_i - \bar{X}_n \left(\frac{1}{n} \sum_{i=1}^n U_i \right) \\ &\rightarrow_p 0 - \mathbb{E}[X_i] \cdot 0 = 0. \end{aligned}$$

Denominator converges to $\text{Var}(X_i)$

- The sample variance can be written as

$$\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X}_n)^2 = \frac{1}{n} \sum_{i=1}^n X_i^2 - \bar{X}_n^2.$$

- By the LLN, $\frac{1}{n} \sum_{i=1}^n X_i^2 \rightarrow_p \mathbb{E}[X_i^2]$ and $\bar{X}_n \rightarrow_p \mathbb{E}[X_i]$.
- By Slutsky's Lemma, $\bar{X}_n^2 \rightarrow_p (\mathbb{E}[X_i])^2$.
- Thus,

$$\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X}_n)^2 = \frac{1}{n} \sum_{i=1}^n X_i^2 - \bar{X}_n^2 \rightarrow_p \mathbb{E}[X_i^2] - (\mathbb{E}[X_i])^2 = \text{Var}(X_i).$$

Multiple regression

- Under similar conditions to 1–4, one can establish consistency of OLS for the multiple linear regression model:

$$Y_i = \beta_0 + \beta_1 X_{1,i} + \dots + \beta_k X_{k,i} + U_i,$$

where $\mathbb{E}[U_i] = 0$.

- The key assumption is that the errors and regressors are uncorrelated:

$$\mathbb{E}[X_{1,i} U_i] = \dots = \mathbb{E}[X_{k,i} U_i] = 0.$$

Omitted variables and OLS inconsistency

- Suppose that the true model has two regressors:

$$\begin{aligned} Y_i &= \beta_0 + \beta_1 X_{1,i} + \beta_2 X_{2,i} + U_i, \\ \mathbb{E}[X_{1,i} U_i] &= \mathbb{E}[X_{2,i} U_i] = 0. \end{aligned}$$

- Suppose that the econometrician includes **only** X_1 in the regression when estimating β_1 :

$$\begin{aligned}
\tilde{\beta}_{1,n} &= \frac{\sum_{i=1}^n (X_{1,i} - \bar{X}_{1,n}) Y_i}{\sum_{i=1}^n (X_{1,i} - \bar{X}_{1,n})^2} \\
&= \frac{\sum_{i=1}^n (X_{1,i} - \bar{X}_{1,n}) (\beta_0 + \beta_1 X_{1,i} + \beta_2 X_{2,i} + U_i)}{\sum_{i=1}^n (X_{1,i} - \bar{X}_{1,n})^2} \\
&= \beta_1 + \beta_2 \frac{\sum_{i=1}^n (X_{1,i} - \bar{X}_{1,n}) X_{2,i}}{\sum_{i=1}^n (X_{1,i} - \bar{X}_{1,n})^2} \\
&\quad + \frac{\sum_{i=1}^n (X_{1,i} - \bar{X}_{1,n}) U_i}{\sum_{i=1}^n (X_{1,i} - \bar{X}_{1,n})^2}.
\end{aligned}$$

- Dividing numerator and denominator by n and applying the LLN as before:

- The noise term vanishes:

$$\frac{\frac{1}{n} \sum_{i=1}^n (X_{1,i} - \bar{X}_{1,n}) U_i}{\frac{1}{n} \sum_{i=1}^n (X_{1,i} - \bar{X}_{1,n})^2} \rightarrow_p \frac{\text{Cov}(X_{1,i}, U_i)}{\text{Var}(X_{1,i})} = 0.$$

- The bias term converges:

$$\frac{\frac{1}{n} \sum_{i=1}^n (X_{1,i} - \bar{X}_{1,n}) X_{2,i}}{\frac{1}{n} \sum_{i=1}^n (X_{1,i} - \bar{X}_{1,n})^2} \rightarrow_p \frac{\text{Cov}(X_{1,i}, X_{2,i})}{\text{Var}(X_{1,i})}.$$

- Therefore,

$$\tilde{\beta}_{1,n} \rightarrow_p \beta_1 + \beta_2 \frac{\text{Cov}(X_{1,i}, X_{2,i})}{\text{Var}(X_{1,i})}.$$

- $\tilde{\beta}_{1,n}$ is **inconsistent** unless:

1. $\beta_2 = 0$ (the model is correctly specified).
2. $\text{Cov}(X_{1,i}, X_{2,i}) = 0$ (the omitted variable is **uncorrelated** with the included regressor).

OVB through the composite error

- In this example, the model contains two regressors:

$$\begin{aligned}
Y_i &= \beta_0 + \beta_1 X_{1,i} + \beta_2 X_{2,i} + U_i, \\
\text{E}[X_{1,i} U_i] &= \text{E}[X_{2,i} U_i] = 0.
\end{aligned}$$

- However, since X_2 is not controlled for, it goes into the error term:

$$\begin{aligned}
Y_i &= \beta_0 + \beta_1 X_{1,i} + V_i, \text{ where} \\
V_i &= \beta_2 X_{2,i} + U_i.
\end{aligned}$$

- For consistency of $\tilde{\beta}_{1,n}$ we need $\text{Cov}(X_{1,i}, V_i) = 0$; however,

$$\begin{aligned}
\text{Cov}(X_{1,i}, V_i) &= \text{Cov}(X_{1,i}, \beta_2 X_{2,i} + U_i) \\
&= \text{Cov}(X_{1,i}, \beta_2 X_{2,i}) + \text{Cov}(X_{1,i}, U_i) \\
&= \beta_2 \text{Cov}(X_{1,i}, X_{2,i}) + 0 \\
&\neq 0, \text{ unless } \beta_2 = 0 \text{ or } \text{Cov}(X_{1,i}, X_{2,i}) = 0.
\end{aligned}$$

Part II: Asymptotic Normality

Why do we need asymptotic normality?

- In the previous lectures, we showed that the OLS estimator has an **exact** normal distribution when the **errors are normally distributed**.
 - The same assumption is needed to show that the T statistic has a t -distribution and the F statistic has an F -distribution.
- In this lecture, we argue that even when the errors are **not** normally distributed, the OLS estimator has an **approximately normal distribution** in large samples, provided that some additional conditions hold.
 - This property is used for hypothesis testing: in large samples, the T statistic has a standard normal distribution and the F statistic has a χ^2 distribution (approximately).

Asymptotic normality

- Let W_n be a sequence of random variables indexed by the sample size n .
 - Typically, W_n will be a function of some estimator, such as $W_n = \sqrt{n}(\hat{\beta}_n - \beta)$.
- We say that W_n has an **asymptotically normal** distribution if its **CDF** converges to a **normal CDF**.
- Let W be any random variable with a normal $N(0, \sigma^2)$ distribution and let F denote its CDF. We say that W_n has an asymptotically normal distribution if for all $x \in \mathbb{R}$:

$$F_n(x) = P(W_n \leq x) \rightarrow P(W \leq x) = F(x) \text{ as } n \rightarrow \infty.$$

- We denote this as $W_n \rightarrow_d W$ or $W_n \rightarrow_d N(0, \sigma^2)$.

Convergence in distribution

- Asymptotic normality is an example of convergence in distribution.
- We say that a sequence of random variables W_n converges in distribution to W (denoted as $W_n \rightarrow_d W$) if the CDF of W_n converges to the CDF of W at all points where the CDF of W is continuous.
- Convergence in distribution is convergence of the **CDFs**.

Central Limit Theorem (CLT)

- An example of convergence in distribution is a CLT.
- Let $X_1, \dots, X_n$ be a sample of **iid** random variables such that $E[X_i] = 0$ and $\text{Var}(X_i) = \sigma^2 > 0$ (finite). Then, as $n \rightarrow \infty$,

$$\frac{1}{\sqrt{n}} \sum_{i=1}^n X_i \rightarrow_d N(0, \sigma^2).$$

- $\rightarrow_d$ means that the CDF of the scaled sum converges to the normal CDF: for every x ,

$$P\left(\frac{1}{\sigma\sqrt{n}} \sum_{i=1}^n X_i \leq x\right) \rightarrow \Phi(x) \text{ as } n \rightarrow \infty,$$

where Φ is the standard normal CDF. For large n , the distribution of the scaled sum is approximately normal.

CLT with non-zero mean

- For the CLT we impose 3 assumptions: **(1)** iid; **(2)** Mean zero; **(3)** Finite variance different from zero.
- If $X_1, \dots, X_n$ are iid but $E[X_i] = \mu \neq 0$, then consider $X_i - \mu$. Since $E[X_i - \mu] = 0$, we have

$$\frac{1}{\sqrt{n}} \sum_{i=1}^n (X_i - \mu) \rightarrow_d N(0, \text{Var}(X_i)).$$

Then

$$\begin{aligned} \frac{1}{\sqrt{n}} \sum_{i=1}^n (X_i - \mu) &= \sqrt{n} \frac{1}{n} \sum_{i=1}^n (X_i - \mu) \\ &= \sqrt{n} \left(\frac{1}{n} \sum_{i=1}^n X_i - \frac{1}{n} \sum_{i=1}^n \mu \right) \\ &= \sqrt{n} (\bar{X}_n - \mu). \end{aligned}$$

CLT for the sample average

- From the previous slide:

$$\frac{1}{\sqrt{n}} \sum_{i=1}^n (X_i - \mu) = \sqrt{n} (\bar{X}_n - \mu).$$

- Thus, the CLT can be stated as

$$\sqrt{n} (\bar{X}_n - \mu) \rightarrow_d N(0, \text{Var}(X_i)).$$

- By the LLN,

$$\bar{X}_n - \mu \rightarrow_p 0,$$

and

$$\text{Var}(\sqrt{n} (\bar{X}_n - \mu)) = n \text{Var}(\bar{X}_n) = n \frac{\text{Var}(X_i)}{n} = \text{Var}(X_i).$$

Properties

- Suppose that $W_n \rightarrow_d N(0, \sigma^2)$ and $\theta_n \rightarrow_p \theta$. Then,

$$\theta_n W_n \rightarrow_d \theta N(0, \sigma^2) \stackrel{d}{=} N(0, \theta^2 \sigma^2),$$

and

$$\theta_n + W_n \rightarrow_d \theta + N(0, \sigma^2) \stackrel{d}{=} N(\theta, \sigma^2).$$

- Suppose that $Z_n \rightarrow_d Z \sim N(0, 1)$. Then,

$$Z_n^2 \rightarrow_d Z^2 \equiv \chi_1^2.$$

- If $W_n \rightarrow_d c = \text{constant}$, then $W_n \rightarrow_p c$.

Asymptotic normality of OLS

- Suppose that:
 1. The data $\{(Y_i, X_i) : i = 1, \dots, n\}$ are iid.
 2. $Y_i = \beta_0 + \beta_1 X_i + U_i$, where $E[U_i] = 0$.
 3. $E[X_i U_i] = 0$.
 4. $0 < \text{Var}(X_i) < \infty$.
 5. $0 < E[(X_i - E[X_i])^2 U_i^2] < \infty$ and $0 < E[U_i^2] < \infty$.
- Let $\hat{\beta}_{1,n}$ be the OLS estimator of β_1 . Then,

$$\sqrt{n}(\hat{\beta}_{1,n} - \beta_1) \rightarrow_d N\left(0, \frac{E[(X_i - E[X_i])^2 U_i^2]}{(\text{Var}(X_i))^2}\right).$$

- $V = \frac{E[(X_i - E[X_i])^2 U_i^2]}{(\text{Var}(X_i))^2}$ is called the **asymptotic variance** of $\hat{\beta}_{1,n}$.

Large-sample approximation for OLS

- Let $\overset{a}{\sim}$ denote “approximately in large samples.”
- The asymptotic normality

$$\sqrt{n}(\hat{\beta}_{1,n} - \beta_1) \rightarrow_d N(0, V)$$

can be viewed as the following large-sample **approximation**:

$$\sqrt{n}(\hat{\beta}_{1,n} - \beta_1) \overset{a}{\sim} N(0, V),$$

or

$$\hat{\beta}_{1,n} \overset{a}{\sim} N(\beta_1, V/n).$$

Proof: decomposition

Write

$$\hat{\beta}_{1,n} = \beta_1 + \frac{\sum_{i=1}^n (X_i - \bar{X}_n) U_i}{\sum_{i=1}^n (X_i - \bar{X}_n)^2}.$$

Now

$$\hat{\beta}_{1,n} - \beta_1 = \frac{\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X}_n) U_i}{\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X}_n)^2},$$

and

$$\sqrt{n} (\hat{\beta}_{1,n} - \beta_1) = \frac{\frac{1}{\sqrt{n}} \sum_{i=1}^n (X_i - \bar{X}_n) U_i}{\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X}_n)^2}.$$

Proof: combining the limits

$$\sqrt{n} (\hat{\beta}_{1,n} - \beta_1) = \frac{\frac{1}{\sqrt{n}} \sum_{i=1}^n (X_i - \bar{X}_n) U_i}{\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X}_n)^2}.$$

In Part I, we established

$$\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X}_n)^2 \rightarrow_p \text{Var}(X_i).$$

We will show that

$$\frac{1}{\sqrt{n}} \sum_{i=1}^n (X_i - \bar{X}_n) U_i \rightarrow_d N(0, \text{E}[(X_i - \text{E}[X_i])^2 U_i^2]),$$

so that

$$\begin{aligned} \sqrt{n} (\hat{\beta}_{1,n} - \beta_1) &= \frac{\frac{1}{\sqrt{n}} \sum_{i=1}^n (X_i - \bar{X}_n) U_i}{\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X}_n)^2} \\ &\rightarrow_d \frac{N(0, \text{E}[(X_i - \text{E}[X_i])^2 U_i^2])}{\text{Var}(X_i)} \\ &\stackrel{d}{=} N\left(0, \frac{\text{E}[(X_i - \text{E}[X_i])^2 U_i^2]}{(\text{Var}(X_i))^2}\right). \end{aligned}$$

Proof: numerator CLT

$$\begin{aligned} \frac{1}{\sqrt{n}} \sum_{i=1}^n (X_i - \bar{X}_n) U_i &= \frac{1}{\sqrt{n}} \sum_{i=1}^n (X_i - \text{E}[X_i] + \text{E}[X_i] - \bar{X}_n) U_i \\ &= \frac{1}{\sqrt{n}} \sum_{i=1}^n (X_i - \text{E}[X_i]) U_i + (\text{E}[X_i] - \bar{X}_n) \frac{1}{\sqrt{n}} \sum_{i=1}^n U_i. \end{aligned}$$

We have

$$\text{E}[(X_i - \text{E}[X_i]) U_i] = \text{E}[X_i U_i] - \text{E}[X_i] \text{E}[U_i] = 0,$$

and $0 < \text{E}[(X_i - \text{E}[X_i])^2 U_i^2] < \infty$, so by the CLT,

$$\frac{1}{\sqrt{n}} \sum_{i=1}^n (X_i - \text{E}[X_i]) U_i \rightarrow_d N(0, \text{E}[(X_i - \text{E}[X_i])^2 U_i^2]).$$

Proof: second term vanishes

It is left to show that

$$(\mathbb{E}[X_i] - \bar{X}_n) \frac{1}{\sqrt{n}} \sum_{i=1}^n U_i \rightarrow_p 0.$$

We have $\mathbb{E}[U_i] = 0$ and $0 < \mathbb{E}[U_i^2] < \infty$. By the CLT,

$$\frac{1}{\sqrt{n}} \sum_{i=1}^n U_i \rightarrow_d N(0, \mathbb{E}[U_i^2]).$$

By the LLN,

$$\mathbb{E}[X_i] - \bar{X}_n \rightarrow_p 0.$$

Hence, the result follows.

Part III: Asymptotic Variance

Asymptotic variance

- In Part II, we showed that when the data are iid and the regressors are exogenous,

$$\begin{aligned} Y_i &= \beta_0 + \beta_1 X_i + U_i, \\ \mathbb{E}[U_i] &= \mathbb{E}[X_i U_i] = 0, \end{aligned}$$

the OLS estimator of β_1 is asymptotically normal:

$$\begin{aligned} \sqrt{n}(\hat{\beta}_{1,n} - \beta_1) &\rightarrow_d N(0, V), \\ V &= \frac{\mathbb{E}[(X_i - \mathbb{E}[X_i])^2 U_i^2]}{(\text{Var}(X_i))^2}. \end{aligned}$$

- For hypothesis testing, we need a **consistent** estimator of the asymptotic variance V :

$$\hat{V}_n \rightarrow_p V.$$

Simplifying V under homoskedasticity

- Assume that the errors are **homoskedastic**:

$$\mathbb{E}[U_i^2 | X_i] = \sigma^2 \text{ for all } X_i \text{'s.}$$

- In this case, the asymptotic variance can be simplified using the Law of Iterated Expectation:

$$\begin{aligned} \mathbb{E}[(X_i - \mathbb{E}[X_i])^2 U_i^2] &= \mathbb{E}[\mathbb{E}[(X_i - \mathbb{E}[X_i])^2 U_i^2 | X_i]] \\ &= \mathbb{E}[(X_i - \mathbb{E}[X_i])^2 \mathbb{E}[U_i^2 | X_i]] \\ &= \mathbb{E}[(X_i - \mathbb{E}[X_i])^2 \sigma^2] \\ &= \sigma^2 \mathbb{E}[(X_i - \mathbb{E}[X_i])^2] = \sigma^2 \text{Var}(X_i). \end{aligned}$$

Estimating V : method of moments

- Thus, when the errors are homoskedastic with $E[U_i^2] = \sigma^2$,

$$V = \frac{E[(X_i - E[X_i])^2 U_i^2]}{(\text{Var}(X_i))^2} = \frac{\sigma^2 \text{Var}(X_i)}{(\text{Var}(X_i))^2} = \frac{\sigma^2}{\text{Var}(X_i)}.$$

- Let $\hat{U}_i = Y_i - \hat{\beta}_{0,n} - \hat{\beta}_{1,n}X_i$, where $\hat{\beta}_{0,n}$ and $\hat{\beta}_{1,n}$ are the OLS estimators of β_0 and β_1 .
- A consistent estimator for the asymptotic variance can be constructed using the **Method of Moments**:

$$\begin{aligned}\hat{\sigma}_n^2 &= \frac{1}{n} \sum_{i=1}^n \hat{U}_i^2, \\ \widehat{\text{Var}}(X_i) &= \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X}_n)^2, \text{ and} \\ \hat{V}_n &= \frac{\hat{\sigma}_n^2}{\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X}_n)^2}.\end{aligned}$$

Why the LLN does not apply directly

- From the previous slide:

$$\begin{aligned}\hat{V}_n &= \frac{\hat{\sigma}_n^2}{\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X}_n)^2}, \\ \hat{\sigma}_n^2 &= \frac{1}{n} \sum_{i=1}^n \hat{U}_i^2, \\ \hat{U}_i &= Y_i - \hat{\beta}_{0,n} - \hat{\beta}_{1,n}X_i.\end{aligned}$$

- When proving the consistency of OLS (Part I), we showed that

$$\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X}_n)^2 \rightarrow_p \text{Var}(X_i),$$

and to establish $\hat{V}_n \rightarrow_p V$, we need to show that $\hat{\sigma}_n^2 \rightarrow_p \sigma^2$.

- The LLN **cannot** be applied directly to

$$\frac{1}{n} \sum_{i=1}^n \hat{U}_i^2$$

because the $\hat{U}_i$'s are not iid: they are **dependent** through $\hat{\beta}_{0,n}$ and $\hat{\beta}_{1,n}$.

Proof: $\hat{\sigma}_n^2 \rightarrow_p \sigma^2$

- First, write

$$\begin{aligned}\hat{U}_i &= Y_i - \hat{\beta}_{0,n} - \hat{\beta}_{1,n}X_i \\ &= (\beta_0 + \beta_1 X_i + U_i) - \hat{\beta}_{0,n} - \hat{\beta}_{1,n}X_i \\ &= U_i - (\hat{\beta}_{0,n} - \beta_0) - (\hat{\beta}_{1,n} - \beta_1)X_i.\end{aligned}$$

- Now,

$$\hat{\sigma}_n^2 = \frac{1}{n} \sum_{i=1}^n \hat{U}_i^2 = \frac{1}{n} \sum_{i=1}^n (U_i - (\hat{\beta}_{0,n} - \beta_0) - (\hat{\beta}_{1,n} - \beta_1)X_i)^2.$$

Completing the consistency proof

- We have

$$\begin{aligned}\hat{\sigma}_n^2 &= \frac{1}{n} \sum_{i=1}^n \left(U_i - (\hat{\beta}_{0,n} - \beta_0) - (\hat{\beta}_{1,n} - \beta_1) X_i \right)^2 \\ &= \frac{1}{n} \sum_{i=1}^n U_i^2 + (\hat{\beta}_{0,n} - \beta_0)^2 + (\hat{\beta}_{1,n} - \beta_1)^2 \frac{1}{n} \sum_{i=1}^n X_i^2 \\ &\quad - 2(\hat{\beta}_{0,n} - \beta_0) \frac{1}{n} \sum_{i=1}^n U_i - 2(\hat{\beta}_{1,n} - \beta_1) \frac{1}{n} \sum_{i=1}^n U_i X_i \\ &\quad + 2(\hat{\beta}_{0,n} - \beta_0)(\hat{\beta}_{1,n} - \beta_1) \frac{1}{n} \sum_{i=1}^n X_i.\end{aligned}$$

- By the LLN,

$$\frac{1}{n} \sum_{i=1}^n U_i^2 \rightarrow_p \mathbb{E}[U_i^2] = \sigma^2.$$

- Because $\hat{\beta}_{0,n}$ and $\hat{\beta}_{1,n}$ are consistent,

$$\hat{\beta}_{0,n} - \beta_0 \rightarrow_p 0 \text{ and } \hat{\beta}_{1,n} - \beta_1 \rightarrow_p 0.$$

Using s^2 instead of $\hat{\sigma}_n^2$

- Thus, when the errors are homoskedastic,

$$\hat{V}_n = \frac{\hat{\sigma}_n^2}{\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X}_n)^2}, \text{ with } \hat{\sigma}_n^2 = \frac{1}{n} \sum_{i=1}^n \hat{U}_i^2,$$

is a consistent estimator of $V = \frac{\sigma^2}{\text{Var}(X_i)}$.

- Similarly,

$$s^2 = \frac{1}{n-2} \sum_{i=1}^n \hat{U}_i^2 \rightarrow_p \sigma^2,$$

and therefore

$$\hat{V}_n = \frac{s^2}{\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X}_n)^2}$$

is also a consistent estimator of $V = \frac{\sigma^2}{\text{Var}(X_i)}$.

- This version has an advantage over the one with $\hat{\sigma}_n^2$: in addition to being consistent, s^2 is also an unbiased estimator of σ^2 if the regressors are strongly exogenous.

Asymptotic approximation

- The result $\sqrt{n}(\hat{\beta}_{1,n} - \beta_1) \rightarrow_d N(0, V)$ is used as the following **approximation**:

$$\hat{\beta}_{1,n} \overset{a}{\sim} N\left(\beta_1, \frac{V}{n}\right),$$

where $\overset{a}{\sim}$ denotes approximately in large samples. Thus, the variance of $\hat{\beta}_{1,n}$ can be taken as approximately V/n .

- With $\hat{V}_n = \frac{s^2}{\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X}_n)^2}$ we have

$$\frac{\hat{V}_n}{n} = \frac{s^2}{\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X}_n)^2} \cdot \frac{1}{n} = \frac{s^2}{\sum_{i=1}^n (X_i - \bar{X}_n)^2}.$$

Connection to the exact result

- From the previous slide:

$$\frac{\hat{V}_n}{n} = \frac{s^2}{\sum_{i=1}^n (X_i - \bar{X}_n)^2}$$

- Thus, in the case of homoskedastic errors we have the following asymptotic approximation:

$$\hat{\beta}_{1,n} \overset{a}{\sim} N \left(\beta_1, \frac{s^2}{\sum_{i=1}^n (X_i - \bar{X}_n)^2} \right).$$

- In finite samples, we have the same result **exactly**, when the regressors are **strongly exogenous** and the errors are **normal**.

Asymptotic T -test

- Consider testing $H_0 : \beta_1 = \beta_{1,0}$ vs $H_1 : \beta_1 \neq \beta_{1,0}$.
- Consider the behavior of the T statistic **under** $H_0 : \beta_1 = \beta_{1,0}$. Since

$$\sqrt{n}(\hat{\beta}_{1,n} - \beta_1) \rightarrow_d N(0, V) \text{ and } \hat{V}_n \rightarrow_p V,$$

we have

$$\begin{aligned} T = \frac{\hat{\beta}_{1,n} - \beta_{1,0}}{\sqrt{\hat{V}_n/n}} &= \frac{\sqrt{n}(\hat{\beta}_{1,n} - \beta_{1,0})}{\sqrt{\hat{V}_n}} \\ &\stackrel{H_0}{=} \frac{\sqrt{n}(\hat{\beta}_{1,n} - \beta_1)}{\sqrt{\hat{V}_n}} \\ &\rightarrow_d \frac{N(0, V)}{\sqrt{V}} \stackrel{d}{=} N(0, 1). \end{aligned}$$

Asymptotic T -test: rejection rule

- Under $H_0 : \beta_1 = \beta_{1,0}$,

$$T = \frac{\hat{\beta}_{1,n} - \beta_{1,0}}{\sqrt{\hat{V}_n/n}} \rightarrow_d N(0, 1),$$

provided that $\hat{V}_n \rightarrow_p V$ (the asymptotic variance of $\hat{\beta}_{1,n}$).

- An asymptotic size α test rejects $H_0 : \beta_1 = \beta_{1,0}$ against $H_1 : \beta_1 \neq \beta_{1,0}$ when

$$|T| > z_{1-\alpha/2},$$

where $z_{1-\alpha/2}$ is a **standard normal** critical value.

- Asymptotically, the variance of the OLS estimator is known; we behave as if the variance were known.

Heteroskedastic errors

- In general, the **errors are heteroskedastic**: $E[U_i^2 | X_i]$ is not constant and changes with X_i .

- In this case, $\hat{V}_n = \frac{s^2}{\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X}_n)^2}$ is **not** a consistent estimator of the asymptotic variance $V = \frac{E[(X_i - E[X_i])^2 U_i^2]}{(\text{Var}(X_i))^2}$:

$$\begin{aligned} \frac{s^2}{\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X}_n)^2} &\rightarrow_p \frac{E[U_i^2]}{\text{Var}(X_i)} \\ &= \frac{\text{Var}(X_i) \cdot E[U_i^2]}{(\text{Var}(X_i))^2} \\ &\neq \frac{E[(X_i - E[X_i])^2 U_i^2]}{(\text{Var}(X_i))^2}. \end{aligned}$$

HC estimator of asymptotic variance

- In the case of heteroskedastic errors, a consistent estimator of $V = \frac{E[(X_i - E[X_i])^2 U_i^2]}{(\text{Var}(X_i))^2}$ can be constructed as follows:

$$\hat{V}_n^{HC} = \frac{\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X}_n)^2 \hat{U}_i^2}{\left(\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X}_n)^2\right)^2}.$$

- One can show that $\hat{V}_n^{HC} \rightarrow_p V$ whether the errors are heteroskedastic or homoskedastic.
- We have the following asymptotic approximation:

$$\hat{\beta}_{1,n} \overset{a}{\sim} N\left(\beta_1, \frac{\hat{V}_n^{HC}}{n}\right),$$

and the standard errors can be computed as $\text{se}(\hat{\beta}_{1,n}) = \sqrt{\hat{V}_n^{HC}/n}$.

HC standard errors in R

- In R, the HC estimator of standard errors can be obtained using the `sandwich` package:

```
library(wooldridge)
library(lmtest)
library(sandwich)
data("wage1")
reg <- lm(wage ~ educ + exper + tenure, data = wage1)
```

- Standard (homoskedastic) standard errors:

```
coeftest(reg)
```

t test of coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-2.872735	0.728964	-3.9408	9.225e-05 ***
educ	0.598965	0.051284	11.6795	< 2.2e-16 ***
exper	0.022340	0.012057	1.8528	0.06447 .
tenure	0.169269	0.021645	7.8204	2.935e-14 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

- HC (robust) standard errors:

```
coeftest(reg, vcov = vcovHC(reg, type = "HC1"))
```


t test of coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	-2.872735	0.807415	-3.5579	0.0004078	***
educ	0.598965	0.061014	9.8169	< 2.2e-16	***
exper	0.022340	0.010555	2.1165	0.0347731	*
tenure	0.169269	0.029278	5.7814	1.277e-08	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1